

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Department of Mechanical Engineering
IM0240 – Machine Design I

Drawing package

Tripteron CNC router machine

Delivery 2 – English cover and drawing index

Group 2

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1. About this package

This package contains the complete set of manufacturing and assembly drawings of the Tripteron CNC router. The drawing sheets that follow this index are the CAD sheets as issued from SolidWorks; they are reproduced unchanged so that the dimensions, tolerances and revision state shown on them remain the ones the parts were made to.

General notes applying to every sheet:

- **All dimensions are in millimetres (mm).** Angles are in degrees.
- Projection: first-angle (ISO-A).
- Sheet format: A3 unless stated otherwise on the sheet.
- Unless otherwise indicated: linear tolerance ± 0.5 mm, angular tolerance $\pm 0.5^\circ$, break all sharp edges.
- Do not scale the drawing; work from the stated dimensions.
- Materials are stated in the title block of each sheet; the structural members are steel tube and plate, the links are $25 \times 25 \times 2$ mm square steel tube.

2. Drawing index

Sheet	Title	Scale	Drawn by
Group 1 – Chassis and structure			
1	Structural table – welded frame	1:10	G. Riaño
2	Plate, 1/8 in – table top	1:5	G. Riaño
3	Top support 1	1:2	G. Riaño
4	Motor support	1:2	G. Riaño
5	Plate, 1/4 in – reinforcement	1:1	G. Riaño
6	Motor-side support	1:1	G. Riaño
7	Auxiliary top support	1:1	G. Riaño
8	Auxiliary support 2	1:1	G. Riaño
Group 2 – Electrical system			
9	Control wiring – microcontroller and stepper drivers	1:1	J. J. García / G. Riaño
10	Wiring diagram, sheet 2	1:1	J. J. García
11	Wiring diagram, sheet 3	1:1	J. J. García
12	Wiring diagram, sheet 4	1:1	J. J. García
13	Wiring diagram, sheet 5	1:1	J. J. García
Group 3 – Guide-supported axis system			
14	Guide system – general assembly	1:3	J. M. Rodríguez
15	Guide system – assembly, second view	1:4	J. M. Rodríguez
16	Bearings, motor and lead screw	1:1	Group 2

Sheet	Title	Scale	Drawn by
17	Guide rail assembly, sheet 1	1:3	G. Riaño
18	Guide rail assembly, sheet 2	1:3	G. Riaño
19	Guide rail assembly, sheet 3	1:3	G. Riaño
20	Guide carriage – detail	2:1	G. Riaño
21	Guide carriage – views	1:1	G. Riaño
22	Bearing support – views	1:1	J. M. Rodríguez
Group 4 – Arms (R-pair system)			
23	R-pair system – component list, qty 3	1:1	R. Bueno
24	Bearing housing tube	2:1	D. Zuluaga
25	Arm link with internal bore, qty 6	2:1	D. Zuluaga
26	Arm component, qty 9	2:1	D. Zuluaga
27	Arm component, qty 3	2:1	D. Zuluaga
28	Arm component, qty 9	2:1	D. Zuluaga
Group 5 – Tool support			
29	Tool (spindle) support	1:1	D. Amell
Group 6 – Assembled machine			
30	Complete machine – general assembly	1:10	G. Riaño
31	Complete machine – views	1:1	D. Amell
32	Washer (spacer)	10:1	D. Zuluaga
33	Internal spacer	5:1	D. Zuluaga
34	Spacer – section A–A	2:1	D. Zuluaga
35	Spacer – views	2:1	D. Zuluaga

The “drawn by” column reproduces the title block of each CAD sheet. Sebastián Rincón Martínez and Juan Manuel Rodríguez Duque carried the static analysis of Delivery 3; Sebastián Rincón also contributed to the concept selection of Delivery 1.

3. Relation to the other deliveries

Group	Where it is verified
Chassis and structure	Delivery 3, Sheet 5: finite element analysis of the welded frame (safety factor 66, maximum deflection 0.102 mm).
Guide-supported axes	Delivery 3, Sheet 3: lead-screw torque, efficiency, stresses and column buckling (safety factor 3.21 against buckling), and Sheet 4: bushing loads and rod deflection of the carriages (safety factor 1.87, the tightest in the machine).
Arms	Delivery 3, Sheet 2: internal loads and bending stress in the six links (maximum 26.1 MPa, safety factor 9.6).
Assembled machine	Delivery 3, Sheet 1: kinematic model, stroke usage and workspace (a 200 mm cube, 8.6 dm ³).
Electrical system	Delivery 1: motor selection (NEMA 34 86HB250, 4.5 N m); Delivery 3, Sheet 3: required torque is 7.1 % of that.

Note on the sheets that follow. The CAD sheets are reproduced as issued in the original submission, so their title blocks are in Spanish and carry the revision data of that submission. The reading key is: *Materia* = course, *Proyecto* = project, *Dibujó* = drawn by, *Verif.* = checked, *Aprob.* = approved, *Escala* = scale, *Unidades* = units, *Cantidad* = quantity, *Formato* = sheet size, *Material* = material, *Hoja 1 de 1* = sheet 1 of 1, *Las cotas se expresan en mm* = all dimensions in millimetres, *No cambie la escala* = do not scale the drawing, *Romper aristas* = break sharp edges.